

ICES SMSF Maturity Scale

(Standardised Macroscopic Scale for Fish Maturity Assessment)

Stage	Possible sub-stages
A. Immature	
B. Developing	Ba. Developing but functionally immature (first-time developer)
	Bb. Developing and functionally mature
C. Spawning	Ca. Actively spawning
	Cb. Spawning capable
D. Regressing/ Regenerating	Da. Regressing
	Db. Regenerating
E. Omitted spawning	
F. Abnormal	

1. Purpose and Scope

The **SMSF maturity scale** was initially developed during WKASMSF (ICES, 2018), improved over the years at WGBIOP (ICES, 2025) and finalised in the Handbook on maturity staging of marine species (ICES, 2026 in publication). This scale provides a unified framework for classifying the reproductive development of most teleost fish. Some taxa (e.g. elasmobranchs) and species with more specialised reproductive modes (e.g. viviparous or hermaphroditic species) may require adapted or dedicated maturity scales.

SMSF scale is compulsory for data uploading to ICES db. WGBIOP strongly encourages to assess maturity directly using the SMSF scale, but some countries still prefer to use their national maturity scales and translate it into SMSF before uploading. Those countries **MUST** use the conversion [table](#) to SMSF scale provided by WGBIOP, in order to ensure consistency with historical data.

⚠ Macroscopic maturity assessment is reliable only when performed within approximately three months prior to, and during, the spawning season.

2. Substages in Focus

Ba – Developing but functionally immature

First-time developer (virgin) **not contributing** to the current spawning season, thus **excluded** from the SSB estimation.

Bb – Developing and functionally mature

First or repeat spawner **contributing** to the current spawning season, thus **included** in the SSB estimation

Ca – Actively spawning

Hydrated oocytes; running (releasing eggs or sperm).

Cb – Spawning capable

- *Batch spawners*: between batches (interbatch interval).

- *Total spawners*: shortly before spawning.

E – Omitted spawning

Fish that have spawned previously (thus not substage Ba, as this fish is developing for the first time) but show no advanced oocyte development (still regenerating Db or early Bb) in the expected period, based on the known oocyte developing time (Rideout & Tomkiewicz, 2011).

3. Why Substages Matter

Accurate distinction between **Ba** and **Bb** is critical for estimating maturity ogives and Spawning Stock Biomass (SSB). Misclassification can lead to biased maturity estimates and affect stock assessment outcomes.

! Countries not uploading substages, particularly Ba and Bb, have the responsibility to provide information to the respective stock coordinators/assessors for each stock, clarifying how their stage B should be interpreted. Specifically, they should indicate whether this stage is to be included in or excluded from the estimation of maturity ogives and in turn Spawning Stock Biomass (SSB).

4. Key Diagnostic Features

Substages	Macroscopical	Histological
A. Immature	No signs of development	Only oogonia and primary growth oocytes
Ba. Developing but functionally immature (first-time developer)	No visible oocytes	Cortical alveoli oocytes
Bb. Developing and functionally mature	Visible oocytes and grainy appearance	Vitellogenic oocytes
Ca. Active spawning	Hydrated oocytes and running	Hydrated oocytes and ovulated eggs
Cb. Spawning capable	Hydrated oocytes (<u>very few or none if inter-batch or total spawner prior to spawning</u>) but blood vessels really pronounced	Hydrated oocytes (<u>very few or none if inter-batch or total spawner prior to spawning</u>) and visible new post ovulatory follicles (POFs)
Da. Regressing	Ovaries contracted, flaccid and <u>some</u> mainly hydrated oocytes	Some hydrated oocytes and POFs
Db. Regenerating	Same as Bb BUT grey cast and thicker wall	POFs and previtellogenic oocytes
E. Omitted spawning	Ovaries with no advanced oocyte development in the expected period	<i>Histological characteristics can vary from species to species</i>
F. Abnormal	Infection; intersex - both female and male gonadal tissues can be recognised; stone-roo (filled with connective tissue); one lobe degenerated; other.	Infection; intersex (both female and male tissues can be recognised); one lobe degenerated; other.

5. Essential background knowledge for Maturity Stagers

To apply the scale correctly, the assessor should know:

1. The **reproductive period** of the assessed stock
2. The **reproductive strategy** of the species (e.g. total spawner or batch spawner)
3. The **oocyte developing time**, i.e. the duration from early oocyte formation to the production of a mature, ovulable egg.

WGBIOP has collated all this information on a stock basis in a [table](#).

6. Guiding Questions for Maturity Stagers

- Look at the whole sample. Are there **running or regressing** individuals in the sample?
 - Has this fish **spawned previously**?
 - Is this fish **likely to contribute** to the current spawning season?
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7. Additional/Alternative use of maturity stages

Above the use of the different substages is explained when the intended purpose is the calculation of maturity ogives—that is, the proportion of mature individuals within a population, at a given age or size, that will contribute to the upcoming spawning season and thus to the next generation.

However, if the objective is to estimate, for instance, size at maturity, all individuals that have reached functional maturity should be considered, even if they will not contribute to the next spawning season.

Accordingly, when the intended scope is **the estimation of the mature population for the current year**, the stages should be classified as follows:

Immature: A

Maturing/Mature: Ba, Bb, Ca, Cb, Da, Db, E, and F

Individuals in substage Ba (outside the three months prior to or during the spawning season), in stage E (omitted spawning), or in stage F (abnormal development) are considered maturing or mature but, for various reasons, will not reproduce during the current year. In this context, these stages should still be included among the maturing/mature individuals used for the maturity ogive.

8. References

- ICES (2018). WKASMSF - Report of the Workshop for Advancing Sexual Maturity Staging in Fish (WKASMSF). ICES Expert Group reports (until 2018). Report. <https://doi.org/10.17895/ices.pub.19212915.v1>
 - ICES (2025). Working Group on Biological Parameters (WGBIOP). ICES Scientific Reports. 7:117. 68 pp. <https://doi.org/10.17895/ices.pub.30940826>
 - Rick M. Rideout, Jonna Tomkiewicz, Skipped Spawning in Fishes: More Common than You Might Think, Marine and Coastal Fisheries, Volume 3, Issue 1, January 2011, Pages 176–189, <https://doi.org/10.1080/19425120.2011.556943>
 - Vitale, F., Maxwell, D. and Follesa, M.C. (Eds) (2026). Handbook on Maturity Staging of Marine Species. ICES Cooperative Research Report (*in publication*)
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9. Contact

[ICES Working Group on Biological Parameters \(WGBIOP\)](#)

For any questions, please post on our GitHub repository: <https://github.com/orgs/ices-eg/projects/38>